

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2024 Course: B. Tech. Branch :Mechanical Engineering & Allied Semester : III Subject Code & Name: BTMC302 Fluid Mechanics Max Marks: 60 Date: 02/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	If 5 m³ of certain oil weighs 40 KN, calculate the specific weight, mass density and specific gravity of this oil.	Apply/05	6
B)	Define the following Pascal's law, Hydrostatic law, total pressure, center of pressure.	Understand- ing/01	6
C)	Explain the conditions of stability for the floating bodies with neat sketches.	Understand- ing/01	6
Q.2	Solve Any Two of the following.		12
A)	Explain the following flows with example Steady and unsteady flow, Uniform and non uniform flow	Understand- ing/02	6
B)	Write a short note on the pitot tube and derive the equation for the velocity of the fluid by using a pitot tube.	Apply/04	6
C)	A pipe through which water is flowing is having diameters 40cm and 20cm at the cross sections 1 and 2 respectively. The velocity of the water at section 1 is 5 m/s. Find velocity head at section 1 and 2 and also rate of discharge.	Apply/05	6
Q. 3	Solve Any Two of the following.		12
A)	A viscous flow is taking place in a pipe of diameter 100mm. The maximum velocity is 2m/s. Find the mean velocity and the radius at which it occurs. Also calculate the velocity at 30 mm from the wall of the pipe.	Apply/05	6
B)	Write a short note on major and minor losses in case of flow through pipe.	Understand- ing/03	6
C)	A compound piping system consists of three pipe segments in series. The	Apply/05	6

	lengths of pipe segment are 1800 m, 1200 m and 600 m and respective diameters of pipe segments are 0.50 m, 0.40 m and of 0.30 m. Find the diameter of a equivalent size pipe having 3600 m length.		
Q.4	Solve Any Two of the following.		12
A)	Explain the concept of separation of boundary layer with a neat sketch.	Understand- ding/03	6
B)	A flat plate 2m×2m moves at 40kmph in a stationary air of density 1.15 kg/m³. If Coefficients of drag and lift are 0.15 and 0.75 respectively, find drag force, lift force and resultant force on the plate and orientation of resultant force.	Apply/05	6
C)	Explain the methods of boundary layer control with neat sketches.	Understand- ding/03	6
Q. 5	Solve Any Two of the following.		12
A)	Write a short note on dimensional analysis and dimensional homogeneity.	Understand- ding/03	6
B)	Explain the Rayleigh's method of dimensional analysis.	Understand- ding/03	6
C)	Define Reynolds number and derive its equation.	Understand- ding/03	6
	*** End ***		